SOHO Network Design and Configuration

1. Network Diagram

[Internet/Broadband] | Wired | [Wi■Fi Router] / | | \ Wireless Wireless Wireless Wired | | | | [Laptop-1][Laptop-2][Smart TV][Printer]

Connection Types: Router to Internet = Wired. Laptops and Smart TV = Wireless. Printer = Wired.

2. Hardware Components

Component	Suggested Model
Broadband Connection	JioFiber/Airtel Xstream Fiber
Wi■Fi Router	TP-Link Archer C6 AC1200
Laptop 1	HP 15s
Laptop 2	Dell Inspiron 15
Smart TV	Xiaomi Smart TV X Series
Printer	HP Smart Tank 580
Ethernet Cable	Cat6 RJ45 Cable
Switch (Optional)	TP-Link LS1005G 5-Port
Wi■Fi Extender (Optional)	TP-Link RE205

3. Router Configuration Steps

• Connect broadband modem/ONT to router WAN port.

• Connect a laptop to router via Wi■Fi or Ethernet.

• Open browser and enter router IP (e.g., 192.168.1.1).

• Log in using default credentials from router label/manual.

• Change SSID to a unique name such as SOHO_Office_Network.

• Select WPA2-PSK security.

• Create a strong Wi■Fi password.

• Change default administrator password.

• Enable DHCP server.

• Save settings and reboot router.

4. IP Addressing Plan

Device	MAC Address	IP Address
Router	AA:BB:CC:DD:EE:01	192.168.1.1
Laptop-1	AA:BB:CC:DD:EE:11	DHCP (192.168.1.101)
Laptop-2	AA:BB:CC:DD:EE:12	DHCP (192.168.1.102)
Smart TV	AA:BB:CC:DD:EE:21	192.168.1.20
Printer	AA:BB:CC:DD:EE:31	192.168.1.30

DHCP Range: 192.168.1.100-192.168.1.200

5. Troubleshooting Laptop Internet Connectivity

Cause 1: Wrong Wi■Fi password. Fix: Re-enter correct password and reconnect.

Cause 2: DHCP/IP issue. Fix: Run ipconfig /renew or reconnect to obtain a valid IP.

Cause 3: Weak signal or router issue. Fix: Move closer to router, restart router, check LEDs.

Cause 4: DNS problem. Fix: Use Google DNS 8.8.8.8 and test connectivity.